

Module 04: CSMA with Collision Detection (CSMA/CD)

AKTU BCS603 Exam Core: "Listen While Talk" mechanism, $T_t \geq 2T_p$ derivation, Ethernet 64-byte minimum frame size origin, and Binary Exponential Backoff.

CSMA/CD: "Listen While Talk", Condition $T_t \geq 2T_p$ & Minimum Frame Size Derivation

1. Collision Detection Operation

- Core Principle: "Listen While Transmit"**
Station continuously compares transmitted signal with received voltage.

Normal Signal: $V_{channel} \approx V_{trans}$

Collision Detected: $V_{channel} > 2 \cdot V_{trans}$ (Peak spike!)
- Action on Collision Detection:**
 - Immediately **ABORT** transmission (Saves bandwidth!).
 - Broadcast a 32/48-bit **JAM SIGNAL** to warn all stations.
 - Increment collision counter k ($k = k + 1$).
 - Execute Binary Exponential Backoff algorithm.
 - If $k > 15$: Give up (Report transmission error to upper layer).

Saves immense channel time compared to sending full broken frame!

2. Mathematical Condition: $T_t \geq 2 \cdot T_p$

$$T_t \geq 2 \cdot T_p$$

Transmission Delay MUST be at least Twice Propagation Delay!

- The Worst-Case Collision Detection Scenario:**
 - Host A transmits at $t = 0$.
 - At $t = T_p$, signal reaches far-end Host B.
 - B transmits; collision occurs at $t \approx T_p$.
 - Collision noise takes another T_p to travel back to Host A.
 - Host A detects collision at $t = 2 \cdot T_p$ (Round Trip Time)!**

$$L_{min} = 2 \cdot T_p \cdot \text{Bandwidth (Bits)}$$

If frame finishes transmitting before $2T_p$, sender CANNOT detect collision!

Binary Exponential Backoff Algorithm (BEB) in Ethernet (IEEE 802.3)

Backoff Formula: $\text{Waiting Time} = R \cdot \text{Slot_Time}$ where R is chosen uniformly at random from range: $0 \leq R \leq 2^k - 1$

Range Scaling: 1st Collision ($k=1$): $R \in \{0, 1\}$. 2nd Collision ($k=2$): $R \in \{0, 1, 2, 3\}$. 10th Collision ($k=10$): $R \in \{0 \dots 1023\}$.

Ethernet Slot Time: In 10 Mbps Ethernet: Slot Time = $51.2 \mu s$ = 512 bit times = **64 Bytes (Minimum Frame Size!)**

1. Minimum Frame Size Formula Derivation

A sender must not finish transmitting before the worst-case collision signal returns:

$$T_t \geq 2 \cdot T_p \implies \frac{L}{B} \geq 2 \cdot \frac{d}{v} \implies L_{\text{min}} = 2 \cdot d \cdot B$$

Critical Engineering Implication: Agar network bandwidth badhai jaye (e.g. 10 Mbps to 100 Mbps to 1 Gbps), to same minimum frame size maintain karne ke liye maximum network cable distance d ko proportionally **kam** karna padta hai! Isliye Gigabit Ethernet me collision domain distance sirf 100 meters tak limited hota hai.

2. Binary Exponential Backoff (BEB) Summary

Collision Count (\$k\$)	Contention Window Range (\$2^k - 1\$)	Possible Values for \$R\$	Collision Probability (2 nodes)
1	0 to 1	{0, 1}	0.50 (50%)
2	0 to 3	{0, 1, 2, 3}	0.25 (25%)
3	0 to 7	{0, 1, 2, 3, 4, 5, 6, 7}	0.125 (12.5%)
10 (Max cap)	0 to 1023	1024 discrete choices	~0.00098
> 16	Abort	Frame dropped; Link failure reported	—